

RTC-SDD 2027: A SPEECH DEEFAKE DETECTION CHALLENGE FOR REAL-TIME COMMUNICATION SCENARIOS

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ABSTRACT

Speech deepfake detection aims to determine whether the source of a speech signal is genuine or spoofed. Although substantial progress has been made in recent years, few studies have focused on the challenges of deepfake speech detection in real-time communication scenarios. In such scenarios, speech enhancement, codec and compression performed by communication platforms, as well as jitter and packet loss during network transmission, can destroy the fine-grained artifacts that detection models rely on, posing significant challenges to model robustness. To promote research on speech deepfake detection in real-time communication scenarios, we propose the RTC-SDD Challenge and provide the first dataset constructed through real RTC transmission. The dataset consists of offline and online subsets, where the online data are obtained by transmitting the corresponding offline data through RTC systems. Through this challenge, we hope to encourage the community to focus on speech deepfake detection in realistic communication environments and to promote theoretical and methodological innovation. More details about the challenge are available at <https://junxue-tech.github.io/rtc-sdd-challenge>.

Index Terms— Speech deepfake detection, real-time communication, robustness

1. CALL FOR PARTICIPATION

1.1. Competition description

The rapid development of speech synthesis technologies has significantly improved the naturalness, authenticity, and controllability of synthetic speech. Although these technologies have played an important role in areas such as education, healthcare, and entertainment, their unethical use has also introduced serious security threats. To address this challenge, the research community has organized a number of Speech Deepfake Detection (SDD) challenges, such as the ASVspoof[1, 2, 3] and ADD[4, 5] series. Recent studies have further explored the robustness and generalization of SDD under real-world conditions by constructing more challenging benchmarks through simulation[6] or social media data collection[7].

In real-time communication (RTC) scenarios, speech serves as an important cue for identity authentication and trust establishment. If attackers use speech synthesis tools to generate speech with the voice characteristics of a target speaker for malicious fraud, it will pose severe threats to individual privacy, property security, and societal security. However, existing research has paid limited attention to SDD in RTC scenarios. The few related studies mainly investigate the impact of codec algorithms, transmission bandwidths, and traditional telephone networks[2]. In practice, the speech processing pipeline of current mainstream RTC platforms is far more complex,

where speech is first processed at the sender side through platform-specific enhancement, encoding, and compression, then transmitted over the network, and finally decoded and reconstructed into a speech waveform at the receiver side. This process introduces cascaded and coupled distortions into the speech signal, which can destroy the fine-grained artifacts relied upon by deepfake detection, thereby degrading model performance. Moreover, since the detailed speech processing pipelines and algorithms of most RTC platforms are black boxes, it is difficult to fully reproduce the signal degradation in real communication environments through algorithmic simulation alone. These challenges highlight the need for dedicated benchmarks and more robust methods for SDD in RTC scenarios.

To fill this gap, we construct the first Speech Deepfake Detection dataset based on real RTC transmission[8] and propose the RTC-SDD Challenge. The dataset consists of two parts collected before and after transmission, denoted as offline and online, respectively. It contains over 400 hours of speech, 7 mainstream RTC platforms, 307 speakers, and 10 deepfake generation methods. In addition, we add echo and multiple types of environmental noise to the offline evaluation set and further transmit the data through real RTC platforms, thereby constructing evaluation data that better reflects real-world communication environments. The evaluation set of this challenge is divided into a clean online subset and a noisy online subset, and the final score is computed as the weighted sum of the scores on the two subsets, enabling a comprehensive evaluation of model performance under diverse RTC conditions.

1.2. Guidelines for participants

- 1) Participating teams must complete registration through the official challenge website at <https://junxue-tech.github.io/rtc-sdd-challenge> and provide basic team information, including the team name, member names, contact emails, and organization. After registration, participating teams can download the dataset from <https://huggingface.co/datasets/JunXueTech/RTCFake/tree/main>.
- 2) Training, development and evaluation sets will be provided. In the first stage of the challenge, the progress evaluation set will be released, while the final evaluation set will be made available one week before the submission deadline.
- 3) We provide two baseline models, XLS-R+AASIST[9] and XLS-R+Nes2Net[10]. The baseline code is available at <https://github.com/JunXue-tech/RTC-SDD>.
- 4) Following the announcement of the final rankings, the top five teams will be invited to submit a two-page paper and present their work at ICASSP 2027.

For further inquiries, please contact the organizers via email at rtcdd2027.challenge@gmail.com.

2. DATASET

RTCFake[8] is adopted as the benchmark dataset for this challenge. The training and development sets provide both pre- and post-RTC transmission data, referred to as offline and online data, while the progress and final evaluation sets contain only post-transmission online data. The dataset is constructed in two stages: offline data construction and RTC transmission.

The offline data construction stage follows a common SDD dataset generation pipeline. We first collect real Chinese and English speech from Chinese-Lips[11] and LibriHeavy[12], and then synthesize fake speech using TTS and VC methods, which cover diverse paradigms, including open-source diffusion-based models, flow-matching models, large-scale model architectures, and a commercial API. The resulting data are split into training, development, and evaluation sets. To simulate realistic communication environments and highlight the effects of noise suppression and echo cancellation, we randomly sample a subset of the evaluation data and add six types of interference, including office and coffee noise¹, echo², and three types of environmental noise[13].

Table 1: Configuration of speech generation methods, transmission platforms, and noise conditions.

ID	Type	Generation	ID	Platform	ID	Noise
G01	TTS	F5-TTS[14]	P01	Zoom	S01	Clean
G02	TTS	OpenAudio-S1[15]	P02	QQ	S02	Office
G03	TTS	VOXCPM[16]	P03	Wechat	S03	Coffee
G04	TTS	LLaSA[17]	P04	Dingtalk	S04	Echo
G05	TTS	IndexTTS2[18]	P05	Lark	S05	Rain
G06	TTS	Doubao[19]	P06	Voov	S06	Footsteps
G07	TTS	SparkTTS[20]	P07	Telegram	S07	Keyboard
G08	VC	CosyVoice[21]	—	—	—	—
G09	VC	SeedVC[22]	—	—	—	—
G10	VC	ChatterboxVC[23]	—	—	—	—

During the RTC transmission stage, we transmit the offline speech through real RTC platforms to obtain online data with authentic transmission-induced distortions. Specifically, two PC devices are configured as the sender and receiver, with playback and recording performed using a virtual audio cable³ and OBS Studio⁴. To improve transmission efficiency, original utterances are concatenated into longer audio segments before transmission and then segmented back into individual utterances afterward. In addition, Whisper-large model[24] is applied to transcribe the speech before and after transmission and perform content consistency checking. Online samples with mismatched content are discarded, resulting in paired offline and online samples.

Table 2: Subset configuration statistics of the dataset.

Subset	#Generator	#Platform	#Female	#Male
Train	G01–G04, G08–G09	P01–P02	55	50
Dev	G01–G04, G08–G09	P01–P03	10	11
Progress/Eval	G01–G07, G08–G10	P01–P07	95	86

Finally, we remove excessively long or short utterances, resample all data to 16 kHz, and store them in the pcm_s16le format. The subset configurations, data splits, and detailed statistics are shown in Tables 1, 2 and 3. Our data split has two main characteristics: 1) speakers are disjoint across subsets while maintaining a relatively

¹<https://media.xiph.org/rnoise/data/>

²<https://github.com/CLAD23/CLAD/tree/main>

³<https://vb-audio.com/Voicemeeter/>

⁴<https://obsproject.com/>

balanced gender ratio; and 2) the evaluation set includes generation methods, RTC platforms, and noise types unseen during training. The progress set is split from the evaluation set and follows the same distribution as the final evaluation set, with no overlapping samples.

Table 3: Statistics of real and fake samples under offline and online settings.

	Train	Dev	Progress		Eval	
			Clean	Noisy	Clean	Noisy
<i>Offline</i>						
Real	7463	1452	—	—	—	—
Fake	31197	6287	—	—	—	—
<i>Online</i>						
Real	6890	1397	4104	916	16416	3662
Fake	30235	6068	20349	4910	81395	19642
Total	75785	15204	24453	5826	97811	23304

3. TASK

The RTC-SDD Challenge defines a single-track speech deepfake detection task under RTC conditions. Given an input speech utterance after RTC transmission, participants are required to determine whether it is bonafide or spoofed. The evaluation data are divided into two subsets: a clean online subset and a noisy online subset. The clean subset evaluates model performance under standard RTC transmission, while the noisy subset further considers environmental noise and echo introduced before RTC transmission. Figure 1 illustrates the overall task setting of the RTC-SDD Challenge.

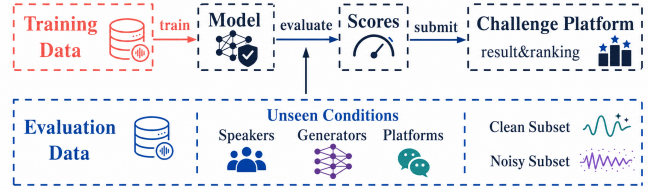


Fig. 1: Overview of the challenge task.

For each evaluation utterance, participants are required to submit a normalized detection score in the range of [0, 1], with a higher score indicating a higher likelihood of being spoofed. The classification threshold is fixed to 0.5 for all submissions.

The official metric is Macro-F1. Let $\mathcal{C} = \{\text{sp}, \text{bf}\}$ denote the class set, where sp and bf represent spoofed and bonafide speech. For each class $c \in \mathcal{C}$, the class-wise F1 score is computed as

$$F1_c = \frac{2TP_c}{2TP_c + FP_c + FN_c}, \quad (1)$$

where TP_c , FP_c , and FN_c denote the numbers of true positives, false positives, and false negatives for class c . Since this is a binary classification task, Macro-F1 is computed as

$$\text{Macro-F1} = \frac{1}{2} (F1_{\text{sp}} + F1_{\text{bf}}). \quad (2)$$

The clean and noisy online subsets are evaluated separately. Let F_{clean} and F_{noisy} denote their Macro-F1 scores. The final score is computed as

$$\text{Macro-F1}_{\text{final}} = \lambda F_{\text{clean}} + (1 - \lambda) F_{\text{noisy}}, \quad (3)$$

where $\lambda = 0.3$. This weighting assigns more importance to the noisy subset, encouraging robust detection under challenging RTC conditions.

4. METHODOLOGY

We provide two baseline systems: XLS-R+AASIST[9] and XLS-R+Nes2Net[10]. Both use a pretrained XLS-R⁵ model to extract frame-level contextual representations, followed by a back-end classifier that produces utterance-level scores. For the back-end classifier, AASIST models spoofing artifacts with graph-based attention, while Nes2Net captures multi-scale discriminative cues from self-supervised speech representations.

During training, each 16 kHz waveform is truncated or zero-padded to 64,600 samples. We optimize the models using Adam with a learning rate of $1e-6$ and a weight decay of $1e-4$. The final checkpoint is selected by the lowest development loss, with early stopping patience set to 10 epochs.

Table 4 reports the Macro-F1 scores of the two baselines on the progress and evaluation sets. Both systems perform well on the clean subset but degrade substantially on the noisy subset, indicating that noisy real-time communication conditions, together with noise suppression and echo cancellation, remain challenging for speech deepfake detection.

Table 4: Baseline results on the progress and evaluation sets (%).

	Progress			Eval		
	Clean	Noisy	Weighted	Clean	Noisy	Weighted
XLS-R+AASIST	91.47	64.36	72.50	90.81	64.38	72.31
XLS-R+Nes2Net	91.01	58.84	68.49	90.80	59.30	68.75

5. ORGANIZATIONAL ASPECTS

5.1. Rules

This section outlines the rules of the challenge, specifying both the required practices and the prohibited actions for participants.

Participation. Participants may join the challenge individually or as a team, but each participant can be affiliated with only one team. The information submitted on the Hugging Face dataset access page should be consistent with the registration form to facilitate the access review.

Data Usage. The official training and development sets can be used for model training, parameter tuning, and model selection. Participants may re-partition or preprocess these subsets as needed. The progress and evaluation sets can only be used to generate prediction scores, and must not be used for training, parameter tuning, model selection, pseudo-labeling, manual annotation, manual analysis.

External Resources. No additional speech data beyond the RTCFake dataset is allowed during training. Participants may use public data augmentation methods or design their own augmentation strategies, but any external data introduced during augmentation must come from publicly available non-speech datasets. Publicly released pretrained model weights are allowed, whereas closed-source model services and unpublished model weights are not permitted.

Model Ensemble Policy. Score-level aggregation or ensemble of multiple models is not allowed. Specifically, participants are not allowed to average, weight, vote, rank-fuse, calibrate, or otherwise post-process and fuse prediction scores from multiple independently

trained models, different model architectures, different random seeds, different training checkpoints, or different systems. Each submission should produce the final prediction scores directly from a single model or a single system.

Submission Policy. The challenge will be evaluated on the Codabench platform. Participants are required to submit a two-column score file in the format of “*id score*”. Each team is allowed to make no more than 30 submissions in total and no more than 3 submissions per day. The final ranking will be determined by the best valid submission.

Report and Reproducibility. Teams invited to submit a final paper are required to provide sufficient details for reproducibility, including the model architecture, training strategy, data augmentation methods, parameter settings, and all external resources used.

All teams must comply with the challenge rules and maintain fairness and integrity throughout the competition.

5.2. Timeline

The important dates of the challenge are as follows:

- **September 1, 2026:** Release of baseline code and dataset; start of the Progress Evaluation Stage.
- **November 9, 2026:** Release of the final evaluation data; start of the Final Evaluation Stage.
- **November 16, 2026:** Final result submission deadline.
- **November 23, 2026:** Release of challenge results and final rankings.
- **December 7, 2026:** 2-page papers due.
- **January 11, 2027:** 2-page paper acceptance notification.
- **January 18, 2027:** Camera-ready 2-page papers due.

Please note that all deadlines are 11:59PM on the respective day as per US Pacific Time.

5.3. Organization

The organizing team consists of members from Wuhan University, China (and other institutions), as listed below.

Yanzhen Ren (Professor at Wuhan University, renyz@whu.edu.cn), primarily working on multimedia content security, including AI interaction security, multimedia forensics, multimedia forgery detection, multimedia information hiding and steganalysis, and multimedia feature representation learning.

Jun Xue (Ph.D. student at Wuhan University, junxue@whu.edu.cn), primarily working on AI-driven speech processing, deepfake detection, and secure human-computer interaction.

Zhuolin Yi (Master’s student at Wuhan University, yizhuolin@whu.edu.cn), primarily working on deepfake detection.

5.4. List of potential participants

⁵https://dl.fbaipublicfiles.com/fairseq/wav2vec/xlsr2_300m.pt

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